

Wahne

1. What is Java programming language? Discuss its key features and advantages.

- Java is a high-level, object-oriented programming language design to have a few implementation dependencies as possible, making it a cornerstone for enterprise and mobile applications. Its key features include platform independence, robust memory management, and runs consistently across various computing environments.

2. Explain the concept of platform independence in Java. How does Java achieve platform independence?

- Platform independence means that Java code can run on any device equipped with a Java Virtual Machine (JVM) without needing recompilation, regardless of the underlying operating system. Java achieves this by compiling source code into an intermediate format called bytecode, which the JVM then executes for the specific host platform.

3. Discuss the main components of a Java program and their roles.

- A typical Java program consists of classes and objects, where the main method serves as the entry point for execution, and the Java compiler (javac) transforms the human-readable source code into executable bytecode. These components work together within the JVM to organize logic into reusable structures, handle data, and manage the flow of the program during runtime.

4. What are the differences between JDK, JRE and JVM in Java?

- The JVM is an engine that executes Java bytecode, the JRE provides the environment and libraries necessary to run Java applications, and the JDK contains the tools, including the compiler and debugger, required to develop them. Think of the JVM as the core engine, the JRE as the complete car ready to drive, and the JDK as the entire factory needed to build the car from the scratch.

5. Explain the difference between interpreted and compiled languages. Is Java an interpreted or compiled language?

~ compiled languages are translated directly into machine code before execution for high performance, whereas interpreted languages are executed line by line by an interpreter at runtime. Java is considered a hybrid language; it is first compiled into bytecode and then interpreted or 'just in time' (JIT) compiled into native machine code by the JVM, combining the benefits of both approaches.